

Investment decisions

Capital budgeting:

It is a financial management tool through which capital expenditure proposals are evaluated.

Factors Influence Capital Budgeting:

- 1) Availability of funds.
- 2) Structure of Capital.
- 3) Taxation Policy
- 4) Govt Policy
- 5) lending Policy.
- 6) Immediate need of the Project.
- 7) Earnings.
- 8) Capital return.

Process of Capital budgeting:

- 1) Project generation.
- 2) Project evaluation.
- 3) Project selection.
- 4) Project initiation - execution.

Types of Risk:

Technical Risk.

Financial Risk.

Economic Risk.

Business Risk.

* Pay back Period :

It refers to period in which the project will generate necessary cash.

$$\text{Pay back Period} = \frac{\text{Original Cost of the Investment}}{\text{Annual Cash In flow}}$$

⇒ Methods of Capital budgeting :

i) Traditional method.

a) Payback Period method.

b) Accounting Rate of Return method.

2) Discounted Cash flow method.

a) Net present value.

b) Internal Rate of Return.

c) Profitability Index.

3) ~~P~~

⇒ Pay back Period Problems :-

1. Mohan and Company is considering the purchase of a machine X & Y each costing ₹ 50000 are available. Cash inflows expected to be follows :

Year	X	Y
1 st yr	15000	5000
2 nd yr	20000	15000
3 rd yr	25000	20000
4 th yr	15000	30000
5 th yr	10000	20000

Calculate Payback Period.

Year	Cash inflow	Cumulative Cash flow
1	15000	15000
2	20000	35000
3	25000	60000

15000

$$PBP = 2 + \frac{15000}{25000}$$

$$= 2.6$$

Year	Cash inflow	Cumulative Cash flow
1	5000	5000
2	15000	20000
3	20000	40000
4	30000	70000

10000

$$PBP = 3 + \frac{10000}{30000}$$

$$= 3.33$$

An industry is considering investing in a project which cost ₹ 1200000. The cashflows are ₹ 240000, ₹ 380000, ₹ 360000, ₹ 400000, & ₹ 500000. Calculate PBP.

Year	Cash inflow	Cumulative Cash flow
1	240000	240000
2	380000	620000
3	360000	980000
4	400000	1380000

320000

$$PBP = 3 + \frac{320000}{400000}$$

$$= 3.8$$

3 A firm has mutually exclusive project X & Y
the details are follows & calculate PBP.

year	X 70000	X 70000
1	10000	60000
2	20000	40000
3	30000	20000
4	45000	10000
5	60000	10000

→ ~~year~~ Project X 70000

year	Cash inflow	Cumulative Cash flow
1	10000	10000
2	20000	30000
3	30000	60000
4	45000	10000

$$PBP = 3 + \frac{10000}{45000}$$

$$= 3.22$$

Project Y 70000

year	Cash inflow	Cumulative Cash flow
1	60000	60000
2	40000	10000

$$PBP = 1 + \frac{10000}{40000}$$

$$= 1.25$$

4. A firm is considering investments in a project which cost ₹ 600000. The cash flows are ₹ 120000, ₹ 140000, ₹ 180000, ₹ 200000 & ₹ 250000. Calculate PBP.

Year	Cash inflow	Cumulative Cash flow.
1	120000	120000
2	140000	260000
3	180000	440000
4	200000	640000

$$PBP = 3 + \frac{160000}{200000}$$

$$= 3.8$$

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Accounting rate of return method:

Also it considers a earnings of the project of the economic life this method is based on conventional accounting concepts.

$$\text{Accounting rate of return} = \frac{\text{Annual Avg net earnings}}{\text{Avg investment}} \times 100$$

where,

$$\text{Annual Avg net earnings} = \frac{\text{total income after deprec \& tax}}{\text{estimated life of machine}}$$

$$\text{Avg investment} = \frac{\text{Original investment} - \text{Scrap value} + WC + SC}{2}$$

WC $\Rightarrow$ Working Capital.

SC $\Rightarrow$ Scrap value.

4. Determine the Aar of sum the following of the 2 machine x & y.

Particular

Original Cost

56125

56125

Additional Investment in working

Capital

5000

6000

Estimated life in year

5yr

5yr

— u — Salvage Scrap value

3000

3000

Income tax

55%

55%

Annual estimated income after 1st year

depreciation & tax

3375

11375

2nd year 5375

9375

3rd year 7375

7375

4th year 9375

5375

5th year 11375

3375

36875

36875

Depreciation as been charged under straightline method.

⇒ Accounting rate of return = $\frac{\text{Annual Avg net Earnings} \times 100}{\text{Average Investment}}$

Annual Avg net Earnings = $\frac{\text{Total Income After Dep & Tax}}{\text{Estimated life of machine}}$

$$= \frac{36875}{5}$$

7375

Avg Investment = $\frac{\text{Original Cost} - \text{Scrap value} + \text{working Capital}}{2}$

$$= \frac{56125 - 3000 + 5000 + 3000}{2}$$

$$= 34563$$

$$ARR = \frac{\text{Annual Avg net Earnings}}{\text{Average Investment}} \times 100$$

$$= \frac{7375}{34563} \times 100$$

$$= 21.33\%$$

$$Y = \text{Annual Avg net Earnings} = \frac{\text{Total Income After Dep \& Tax}}{\text{Estimated life of machine}}$$

$$= \frac{36875}{5}$$

$$= 7375$$

$$\text{Avg Investment} = \frac{\text{Orig Cost} + \text{Scrap value} + \text{WC} + \text{SV}}{2}$$

$$= \frac{56125 - 3000 + 6000 + 3000}{2}$$

$$= 35563$$

$$ARR = \frac{\text{Annual Avg net Earnings}}{\text{Avg Investment}} \times 100$$

$$= \frac{7375}{35563} \times 100$$

$$= 20.7378\%$$

2. Cal the ARR for project A & B from the following.

Particulars	A	B
Investment	20000	30000
Expected life	4yr	5yr

Project net income after Depx & tax

Year	1 st yr	2 nd yr	3 rd yr	4 th yr	5 th yr
A	2000	1500	1500	1000	—
B	3000	3000	2000	1000	1000

which project should be undertaken.

$$\Rightarrow \text{AANE} = \frac{\text{Total Income after Depx \& tax}}{\text{Estimated life}}$$

$$\begin{aligned} A &= \frac{6000}{4} = 1500 \\ B &= \frac{10000}{5} = 2000 \end{aligned}$$

$$\text{Avg Inv} = \frac{\text{Original Cost}}{2}$$

$$\begin{aligned} A &= \frac{20000}{2} = 10000 \\ B &= \frac{30000}{2} = 15000 \end{aligned}$$

$$\text{ARR} = \frac{\text{AANE}}{\text{Avg Inv}} \times 100$$

$$A = \frac{1500}{10000} \times 100 = 15\%$$

$$B = \frac{2000}{15000} \times 100 = 13.33\%$$

Cal the Avg rate of return from the following information.

Particulars	A	B
Investment	250000	300000
Estimated life	4yr	5yr
Scrap value	5000	10000

Annual income after Depx & tax.

1 st	2	3	4	5
4500	5500	8500	9500	-
6000	5000	8000	7000	6000

Determine which project is suitable.

$$AANE = \frac{\text{Total Income ADET}}{\text{Estimated life.}}$$

$$A = \frac{28000}{4} = 7000 \quad B = \frac{32000}{5} = 6400$$

$$\text{Avg Invest} = \frac{\text{Original Cost} - \text{scrap value} + \text{WC} + \text{SV}}{2}$$

$$A = \frac{250000 - 5000 + 0 + 5000}{2} = 127500$$

$$B = \frac{300000 - 10000 + 0 + 10000}{2} = 150000$$

$$ARR = \frac{AANE}{AI} \times 100$$

$$A = \frac{7000}{127500} \times 100 = 5.49\%$$

$$B = \frac{6400}{150000} \times 100 = 4.27\%$$

⇒

Net present value :

NPV method recognises the impact of time value of money. It is considered has best method of evaluating investment proposal.

* Accept @ Reject Criteria :

In case NPV is +ve the project is Selected for investment. If NPV is -ve the project is Rejected.

$$NPV = \text{Present value Cash inflow} - \text{Cash outflow (Investment)}$$

1. The Sintax Company is Planning to purchase a machine to alternative machines are selected for evaluation each cost ₹ 300 000.

The Cash inflows are expected has follows.

Year	Cash inflow 'X'	CIF 'Y'
1 st yr	20000	40000
2 nd yr	65000	110000
3 rd yr	90000	120000
4 th yr	150000	140000
5 th yr	175000	200000

The Company expected Rate of Return is 10%.

Evaluate the profitability of 2 machines by taking

Present value 10%. The PV of ₹ 1 at 10%.

1 st yr	0.909
2 nd yr	0.826
3 rd yr	0.751
4 th yr	0.683
5 th yr	0.621

Machine X

Year	Discount factor	Cash in flow	C = A x B Present Value Cash in flow
1	0.909	20000	18180
2	0.826	65000	53690
3	0.751	90000	67590
4	0.683	150000	102450
5	0.621	175000	108675

Machine Y

Year	Discount factor	Cash in flow	PV Cash in flow
1	0.909	40000	36360
2	0.826	110000	90860
3	0.751	120000	90120
4	0.683	140000	95620
5	0.621	200000	124200

Present value Cash in flow = 350585

Cash out flow = 300000

Net Present value = 50585

PV Cash in flow = 437160

- Cash out flow = 300000

NPV

137160

∴ Machine 'Y' is more suitable compare to machine 'X'.

PQR Ltd is considering the purchase of machinery to alternative machinery 1, 2 are suggested each costing ₹ 500000.

The cash in flow are expected has follows:

Year	1 st	2 nd	3 rd	4 th	5 th
Machine 1	45000	130000	150000	250000	170000
Machine 2	130000	170000	210000	130000	90000
Discount factor 12%	0.893	0.797	0.712	0.636	0.567

Calculate 1) NPV. 2) Profitability Index.

	machine 1				machine 2			
	year	Discount factor	Cash in flow	PV Cash inflow	year	Discount factor	Cash in flow	PV Cash inflow
	1	.893	45000	40185	1	.893	130000	116090
	2	.797	130000	103610	2	.797	170000	135490
	3	.712	150000	106800	3	.712	210000	149520
	4	.636	250000	159000	4	.636	130000	82680
	5	.567	170000	96390	5	.567	90000	51030
	PV Cash inflow			505985	PV Cash inflow			534810
	- Cash out flow			500000	- Cash out flow			500000
	NPV			5985	NPV			34810

Machine 2 has more profitable compare to machine 1.

$$2. \text{ Profitability Index} = \frac{\text{Total PV Cash inflow}}{\text{Investment}} \times 100$$

$$\text{machine 1} = \frac{505985}{500000} \times 100$$

$$101.197\%$$

$$\text{machine 2} = \frac{534810}{500000} \times 100$$

$$106.962\%$$

3 X Company Ltd is considering the purchase of a new machine. The alternative investment for machineries are available (X & Y) each costing ₹ 75000 & cash in flows has flows.

year	1 st	2 nd	3 rd	4 th	5 th
Machine X	30000	30000	20000	10000	5000
Machine Y	40000	30000	20000	10000	10000

The Company expected return on Capital 10%.

Risk premiums are 2% & 7% respectively for investment X & Y which investment should be preferred.

discount factor = $\frac{1}{(1+r)^n}$ $\rightarrow$ for 12% $\frac{1}{(1+0.12)^n}$

for 12% 1st = 0.892
2nd = 0.797
3rd = 0.711
4th = 0.635
5th = 0.567

for 17% 1st = 0.854
2nd = 0.7305
3rd = 0.624
4th = 0.533
5th = 0.456

machine X

year	discount factor (12%)	Cash In flow	PV Cash In flow
1	0.892	30000	26760
2	0.797	30000	23910
3	0.711	20000	14220
4	0.635	10000	6350
5	0.567	5000	2835

~~77880~~
~~75000~~

machine Y

year	discount factor (17%)	Cash In flow	PV Cash In flow
1	0.854	40000	34160
2	0.730	30000	21900
3	0.624	20000	12480
4	0.533	10000	5330
5	0.456	10000	4560

PV Cash In flow = 78430

- Cash out flow = 75000

NPV = 3430

PV Cash In flow = 74075

- Cash out flow = 75000

NPV = -925

machine X will be selected.

Note: Cash inflow refers to EBD&T In case of Pay back Period

& NPV Cash inflow are Earning before Dep & tax Considered for PBP & NPV.

* For ARR After income @ tax Profit after Dep & tax Considered.

2) A Company is considering to purchase a machine & machines are available X & Y Costing ₹ 50000. Earnings after taxation are expected as follows.

year	1 st	2 nd	3 rd	4 th	5 th
M'X	15000	20000	25000	15000	10000
M'Y	5000	15000	20000	30000	20000

Evaluate 2 alternatives according to

A. PB Method B. Return on Investment C. NPV
(Cost of Capital 10%) assume straightline method of depreciation. the discount factors has

1 st	2 nd	3 rd	4 th	5 th
.909	.826	.751	.683	.621

⇒ i) NPV

machine X						machine Y					
year	Profit after Dep & T	Dep	Earning before Dep & T	Dis Factor	PV C Inflow	year	Profit after Dep & T	Dep	Earning before Dep & T	Dis Factor	PV C Inflow
1	15000	10000	25000	.909	22725	1	5000	10000	15000	.909	13635
2	20000	10000	30000	.826	24780	2	15000	10000	25000	.826	20650
3	25000	10000	35000	.751	26285	3	20000	10000	30000	.751	22530
4	15000	10000	25000	.683	17075	4	30000	10000	40000	.683	27320
5	10000	10000	20000	.621	12420	5	20000	10000	30000	.621	18630
PV Cash Inflow					103285	PV Cash Inflow					102765
- Cash Outflow					50000	- Cash outflow					50000
NPV					53285	NPV					52765

2) Pay Back Period:-

Project machine X '50000'			machine Y '50000'		
year	Cash In flow	Cumulative Cash flow	year	Cash In flow	Cumulative Cash flow
	25000	25000	1	15000	15000
	30000	35000	2	25000	40000
	25000	60000	3	30000	70000
			4	30000	100000

Pay Back Period = $1 + \frac{25000}{35000}$

$= 1.714$
 $= 1.71$

Pay Back Period = $2 + \frac{10000}{30000}$

$= 2.33$

Accounting rate of return:-

machine X

Annual Avg net Earnings = $\frac{\text{Total Income After Depreciation \& tax}}{\text{Estimated life of machine}}$

machine X = $\frac{85000}{5}$
 $= 17000$

machine Y = $\frac{90000}{5}$
 $= 18000$

Avg investment = $\frac{\text{Original Cost}}{2}$

machine X & Y = $\frac{50000}{2}$
 $= 25000$

Accounting rate of return = $\frac{\text{Annual Avg net Earnings}}{\text{Avg investment}} \times 100$

machine X = $\frac{17000}{25000} \times 100$
 $= 68\%$

machine Y = $\frac{18000}{25000} \times 100$
 $= 72\%$

5. A Company Considering an investment proposal to instal a new machine the project will cost ₹ 50000 & will have a life of 5yr & no salvage value. Tax rate is 50% straight line method of depre. is follows. Estimated net income before depre. & tax. are:

year	1 st	2 nd	3 rd	4 th	5 th
Net income before depre.	10000	11000	14000	15000	25000

Evaluate project using NPV & at 10% Profitability Index.

year	Earning Before D&T	Discount Factor	PV Cash in flow
1	10000	0.909	9090
2	11000	0.826	9086
3	14000	0.751	10514
4	15000	0.683	10245
5	25000	0.620	15500
Total PV Cash in flow			54435
Cash out flow			50000
NPV			4435

$$\text{Profitability Index} = \frac{\text{Total PV Cash in flow}}{\text{Investment}} \times 100$$

$$= \frac{54435}{50000} \times 100$$

$$= 108.87\%$$

$$= 108.87\%$$

6 A firm has initial investment of ₹ 100000 and cash inflows has follows

year	1 st	2 nd	3 rd	4 th	5 th
Cash inflow	50000	40000	30000	10000	
PV factor 10%	.909	.826	.751	.683	

Find out NPV.

year	Cash inflow	Discount factor	PV Cash inflow
1	50000	.909	45450
2	40000	.826	33040
3	30000	.751	22530
4	10000	.683	6830
PV cash inflow			107850
Cash out flow			100000
NPV			7850

From the following data which project is better for investment

year	Invest	1 st	2 nd	3 rd	4 th
Cash inflow A	15000	7000	6000	2000	
" B	20000	8000	7000	6000	

Adjustment

risk less discount rate is 6%.

Project A is less risky compare to Project B The Management risk premium rate at 6% PA & 10% for Project B.

For discounting Cash inflow.

Discount Rate

Project A = $6 + 6 = 12\%$

Project B = $6 + 10 = 16\%$

Project A 15000

Project B 20000

year	Cash In flow	Discount factor 12% $\frac{1}{(1+0.12)^n}$	PV Cash In flow	year	Cash In flow	Discount factor 16% $\frac{1}{(1+0.16)^n}$	PV Cash In flow
1	7000	0.892	6244	1	8000	0.862	6896
2	6000	0.797	4782	2	7000	0.743	5201
3	2000	0.711	1422	3	6000	0.640	3840
PV Cash In flow =			12448	PV Cash In flow =			15937
- Cash out flow (Inv) =			15000	- Cash out flow =			20000
NPV =			-2552	NPV =			-4063

∴ The firm Project A & B are rejected.

- 5 A Company under Consideration to mutually exclusive Project for increasing its plant capacity. The manager has developed pessimistic, most likely & optimistic estimate of the annual cash in flow associated with each project.

Particulars	X	Y
Net Investment	60000	60000
Cash Inflows		
Pessimistic	2400	7400
most likely	8000	8000
Optimistic	14000	9000

- D Determine the NPV associated with each estimate for both the project. The project have 20 yrs of life & Company Cost of Capital is 10%.
- 2) Which Project do you consider should be selected by the company & why? Present value factor at 10% for 20 years is 8.514.

$$1.01 = 1 + 0.01 = 1.01$$

$$1.02 = 1 + 0.02 = 1.02$$

Project X

	Pessimistic	most likely	Optimistic
Cash Inflow	2400	8000	12000
r factor	8.514	8.514	8.514
Cash Inflow	20434	68112	119196
Cash out flow	60000	60000	60000
NPV	-39566	8112	59196

Project Y

	Pessimistic	most likely	Optimistic
Cash Inflow	7400	8000	9000
r factor	8.514	8.514	8.514
Cash Inflow	63004	68112	76626
Cash out flow	60000	60000	60000
NPV	3004	8112	16626

2/8/23

al Industries is contemplating on buying a new machine for \$50,00,000 with additional working capital of \$10,00,000. The machine is expected to have an economic life of 5 yrs with no salvage value. The firm uses straight line method of deprec and the same expected for tax purpose. The machine is expected to generate an incremental increase in the cash operating income of \$20,00,000 per yr. a period of 5 yrs. Determine tax rate & inflation is expected to be 6% per year. the firm cost of capital. In real terms is 10%. Is the company whether the machine should be purchased. Show your NPV calculation in real terms. working capital requirement will remain unchanged throughout the, unchange.

160. Period.

Operating Income = EBIT

⇒ i) Cash out flow

Cost of machine - 50 00 000

+ working Capital - 10 00 000

Cash out flow - 60 00 000

A Cash inflow after Tax:

Operating Income 20 00 000

- Tax Rate (35%) 7 00 000

13 00 000

X PV factor (5yr @ 10%) 3.79

Annuity 4928300

B Depreciation:

Straight line method $\frac{5000000}{5} = 1000000$

- Tax rate 35% 350000 → Tax saving.

C NPV = Present value Cash inflow - Cash out flow

Year	Tax saving	PV factor 6%	C F A T	Dr @ 10%	PV Cash inflow
1	350000	0.943	330050	0.909	300015.45
2	350000	0.889	31150	0.826	257009.9
3	350000	0.839	293650	0.751	220531.15
4	350000	0.792	277200	0.683	189327.6
5	350000	0.747	261450	0.621	162367.45
Present value				→	1129244.55

D. Working Capital - 10 00 000

DF 6% for 5yr 0.747

747000

DF 10% for 5yr 0.621

163887

Total Present value = 4928300 + 1129245 + 463887

- Cash = 6521432

- Cash outflow = 6000000

NPV = 521432

Company should purchase a machine the NPV is Positive.

Internal rate of return.

3/8/23

$$RR = A - \frac{C-D}{C-D} \times B-A$$

where A = dis factor lower total state

B = dis factor higher total state

C = Pres value Cash in flow lower total state

D = PV Cash in flow higher total state

0 = original cash out-flow.

following are particulars given for two firms

Particulars	A Ltd.	B Ltd.
Initial Investment	500	900
of the first year	100	1269
	200	864
	200	(-1305)
	250	

at 25%.

at 20%.

Rank the projects by NPV and IRR using 15% and discount factors.

1	2	3	4
0.870	0.756	0.658	0.572
0.833	0.694	0.579	0.52

Rank the NPV for A Ltd.

Rank the discount factor & IRR.

Alt

year	Cash Inflow	discount factor 15%	PV Cash Inflow	discount factor 20%	PVCF
1	100	0.870	87	0.833	833.3
2	200	0.756	151	0.694	139
3	200	0.658	132	0.579	116
4	250	0.572	143	0.482	121
			513		459

A) $NPV = 513 - 500 \Rightarrow 13$ (PV - investment)

$NPV = 459 - 500 \Rightarrow 41$

$$IRR = A + \frac{C-D}{C-D} \times (B-A)$$

lower DF $\leftarrow$

$$= 15\% + \frac{513 - 500}{513 - 459} \times (20 - 15)$$

$$= 16.2$$

Alt

year	Cash Inflow	discount factor 15%	PV Cash Inflow	discount factor 20%	PVCF
1	1269	0.870	1104.03	0.833	1057.1
2	864	0.756	653.2	0.694	599.62
3	1305	0.658	858.7	0.579	755.6
4	—	0.572	—	0.482	—
			898.5		1901.12

$NPV = 898.5 - 900 \Rightarrow -1.5$

$NPV = 901.12 - 900 = 1.12$

$$IRR = A + \frac{C-D}{C-D} \times (B-A)$$

$$= 15 + \frac{898.5 - 900}{898.5 - 901.12} \times (20 - 15)$$

$898.5 - 901.12$

A Project is estimated Cost ₹ 16800 It is expected life of 3yr and generate cash inflows ₹ 8000, ₹ 7000 & ₹ 6000 respectively. Calculate IRR Present value ₹ 1 for 3yr.

year	14%	15%
1	• 8772	• 8696
2	• 7695	• 7561
3	• 6750	• 6575

Find out the NPV & IRR:

year	Cashin flow	DF 14%	PV Cashin flow	DF 15%	PVCF
1	8000	• 8772	7018	• 8696	6957
2	7000	• 7695	5387	• 7561	5293
3	6000	• 6750	4050	• 6575	3945
			16455		16195

$$NPV = 16455 - 16800 = 255$$

$$16195 - 16800 = -5$$

$$IRR = A + \frac{C-D}{C-B} \times (B-A)$$

$$= 14 + \frac{16455 - 16800}{16455 - 16195} \times (15 - 14)$$

$$= 14.9807$$